

PORTOFOLIO

Meven MENOU

product design engineer

ABOUT ME

Mechanical Engineer specialized in product design and industrialization, with proven experience delivering innovative products from R&D through to market launch. Strong technical background in manufacturing processes, DFM, 3D CAD, and tolerancing on 2D drawings using SolidWorks.

Currently working at Quad Lock, where I cleaned up 10+ years of CAD data for a PDM migration, established product lifecycle processes, and trained teams on Windchill.

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Personal & Academic PROJECTS

inmotion AIRBAG

Smart airbag vest using an electronic triggering system to enhance user safety across sports applications



MOONBIKES

The world's first electric snowbike, redefining mountain exploration



QUAD LOCK®

Windchill PDM System implementation within QuadLock engineering teams.



Relevant projects from personal, business, and academic experience



WINDCHILL PDM IMPLEMENTATION

Founded in 2011, Quad Lock grew quickly and became a leader in the industry, but the company suffered from historical CAD bad practices and a lack of established lifecycle processes.



BROKEN REFERENCES

DUPLICATE FILES

UNRELIABLE DATABASE & BOMS

MANUAL EXCEL TRACKING

INCONSISTENT NAMING

CLEANED AND MIGRATED 10YEARS+ OF CAD DATA

ANALYSED FILES TO SUPPRES DOUBLON

MASS RENAMED AND PROPERTIES CLEANED

CREATED PDM M-BOM STRUCTURE GENERATED BY CAD-BOM

UPDATED 2D/3D TEMPLATES

ESTABLISHED REVISION AND PROMOTION PROCESS WITH THE PDM

CREATE TRAINING SLIDESHOW AND VIDEOS

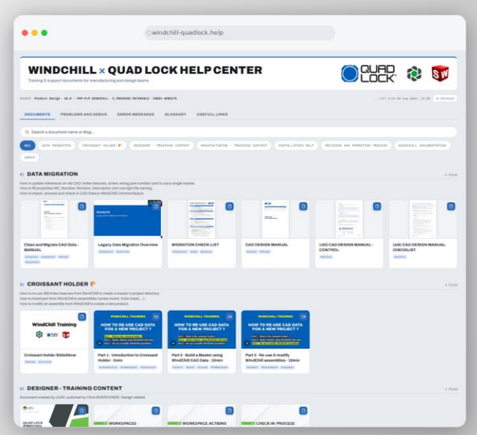
TRAINED DESIGNER AND ENGINEER ON WINDCHILL

CODED A TRAINING INDEX WEBSITE

ONE SINGLE SOURCE OF TRUTH WITH 1600+ ACTIVE PARTS

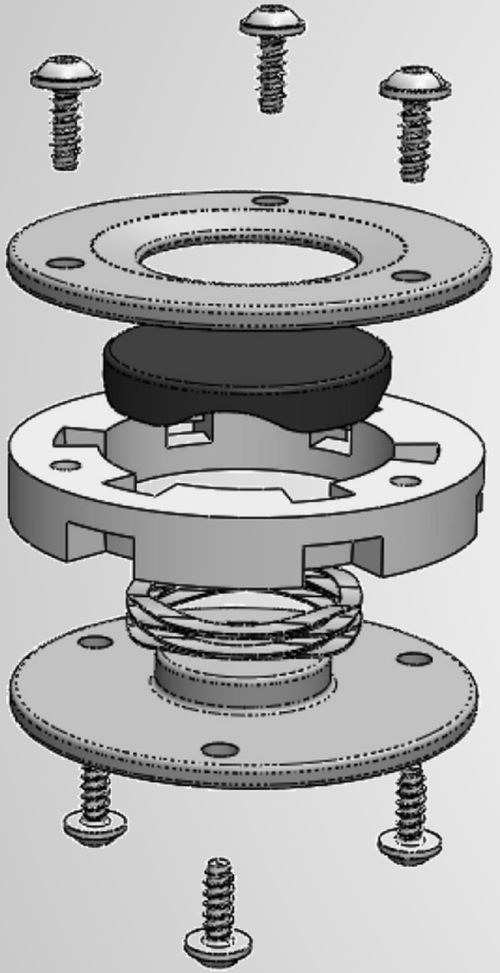
PRODUCT LIFECYCLE PROCESS

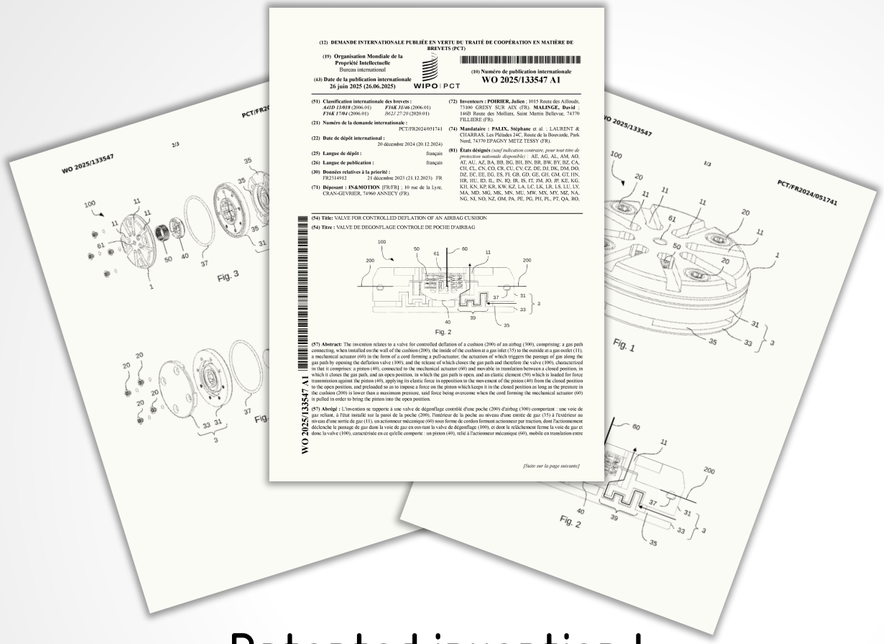
SOLIDWORKS X WINDCHILL TRAINING WEBSITE



DEFLATING VALVE FOR AIRBAG VEST

Brief : Develop a user-friendly mechanical system allowing the airbag to deflate within a few seconds without compromising its protective performance





(12) DEMANDE INTERNATIONALE PUBLIÉE EN VERTU DU TRAITÉ DE COOPÉRATION EN MATIÈRE DE BREVETS (PCT)

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Bureau international

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WO 2025/133547 A1

(54) Title: VALVE FOR CONTROLLED DEFLATION OF AN AIRBAG CUSHION

(54) Titre : VALVE DE DEGONFLAGE CONTROLE DE POCHE D'AIRBAG

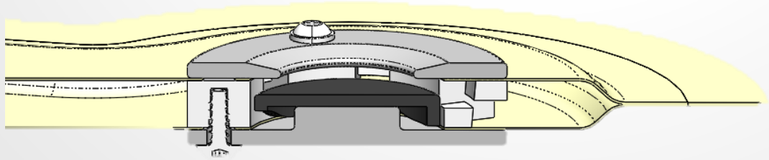
Fig. 1

Fig. 2

Fig. 3

Patented invention !

Click to acces





The airbag fabric is cut to allow the integration of a spring-assisted valve system, which clamps a layer of textile

Product used on since Rally Raid Dakar 2023 and presented at EICMA 2026

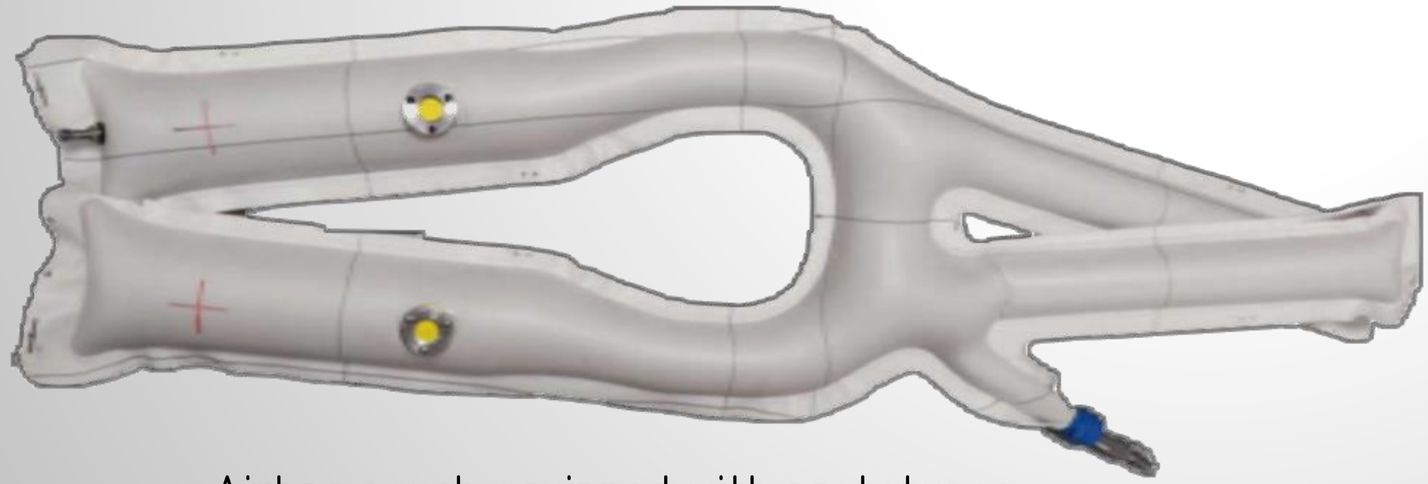
WORK

Prototyping using CNC-machined aluminum alloy components to ensure that the airbag-equipped product complies with road safety regulations, particularly regarding impact test performance, as the cushioning volume is affected.

An initial production batch was launched to successfully validate the product under real conditions during the 2023 and 2024 Dakar Rally.



Impact tests



Airbag pouch equipped with prototypes



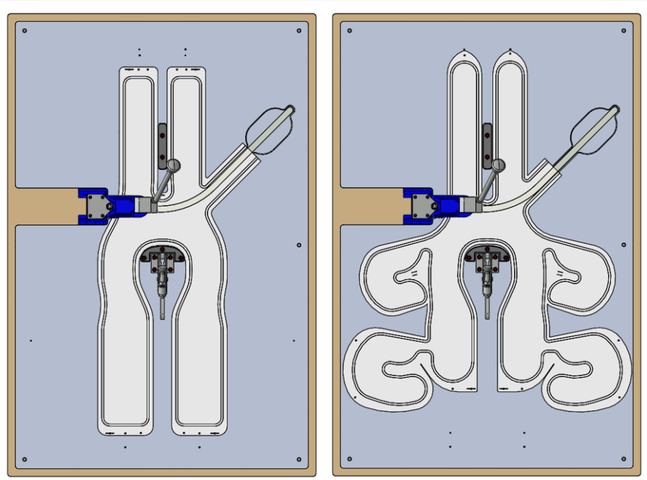
CNC machined version

INDUSTRIALIZATION

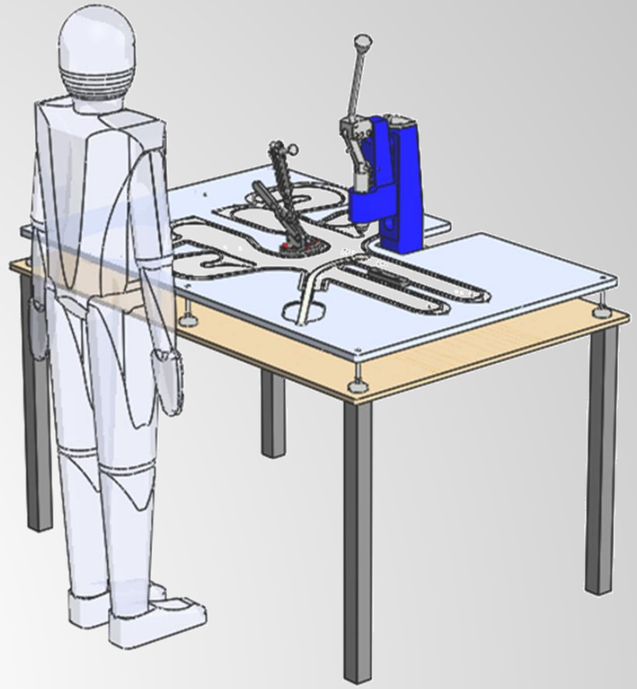
R&D and technical decision-making for the airbag cushion cutting process, using a manual press, specially designed punching tools, and a dedicated workstation.

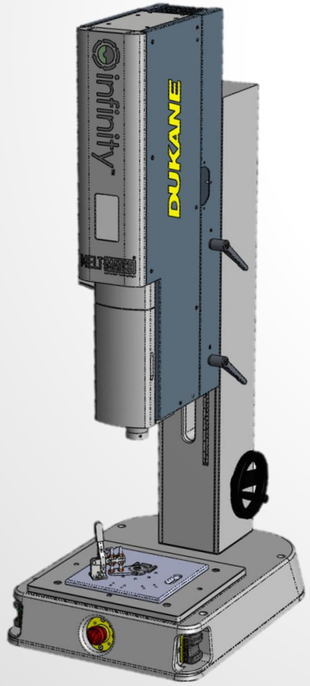
Design of an injection-molded plastic version and validation of an ultrasonic assembly solution after the creation of the ultrasonic horn.

Development of a production tool to ensure repeatability and reliability in ultrasonic welding processes.



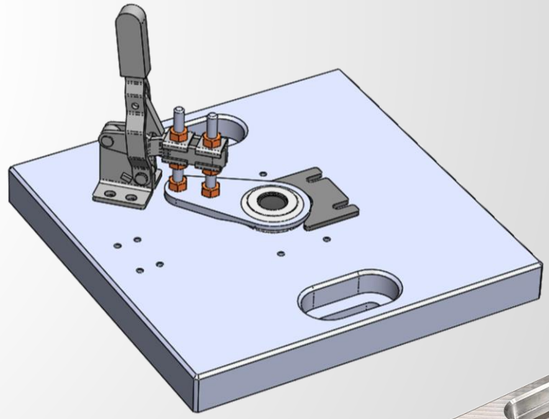
Cuting workstation

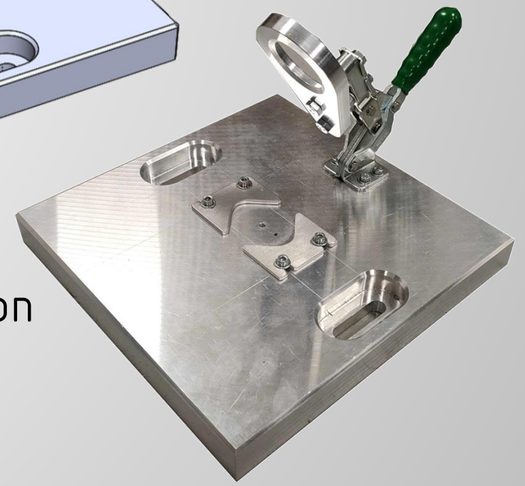






Ultrasonic machine and horn developed



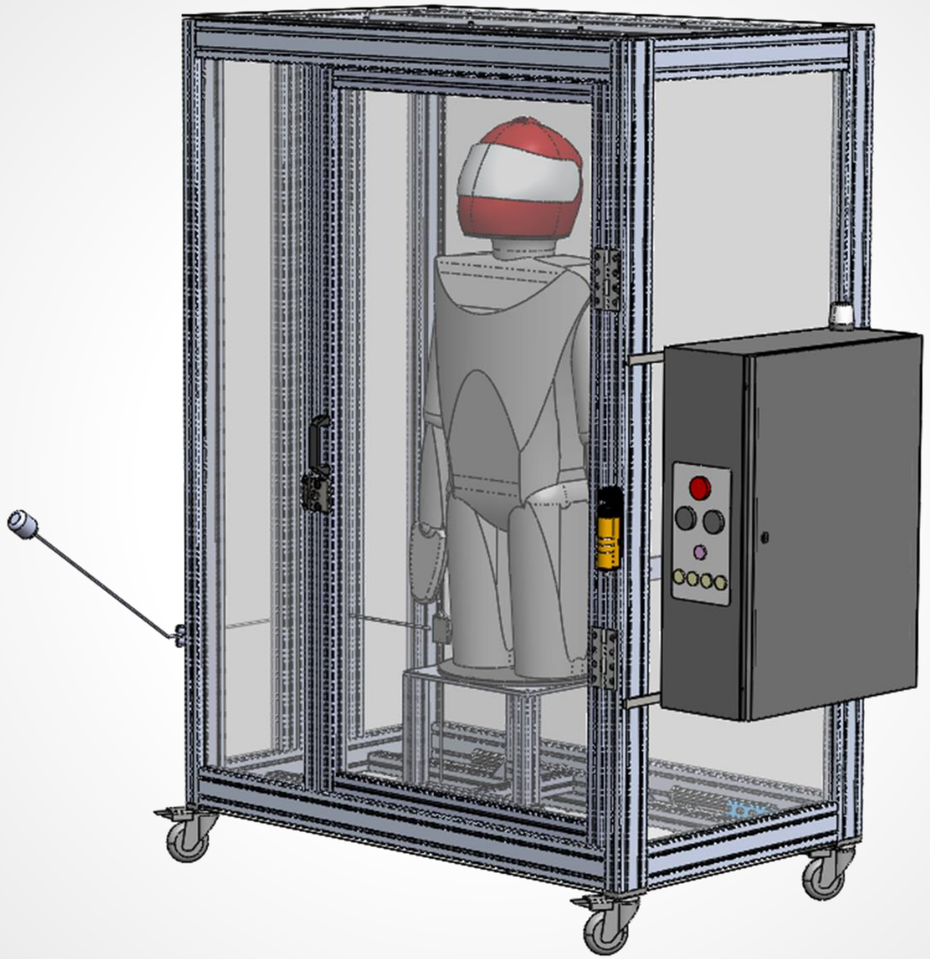


Ultrasonic production tooling to ensure repeatability

PROTECTIVE TEST BENCH

Brief : Design and construction of a protective structure for engineers during airbag prototype testing



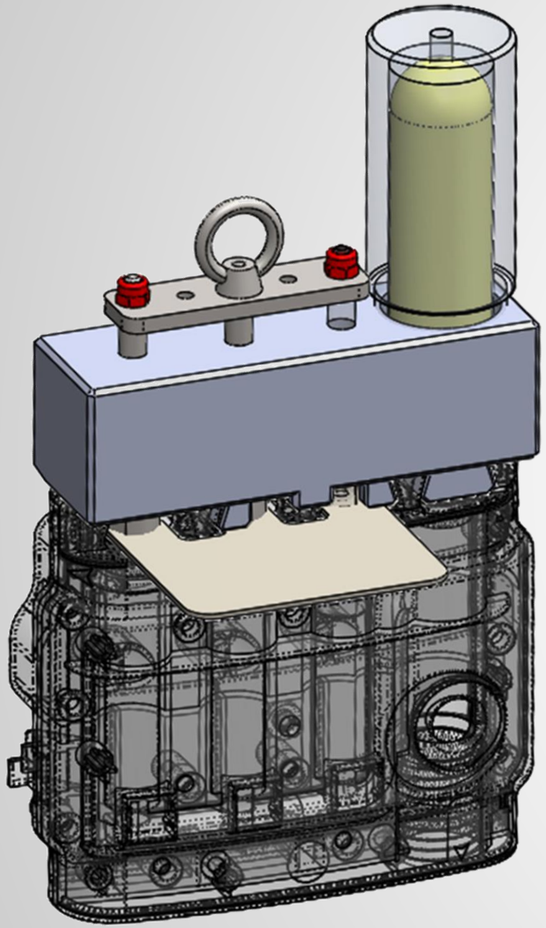




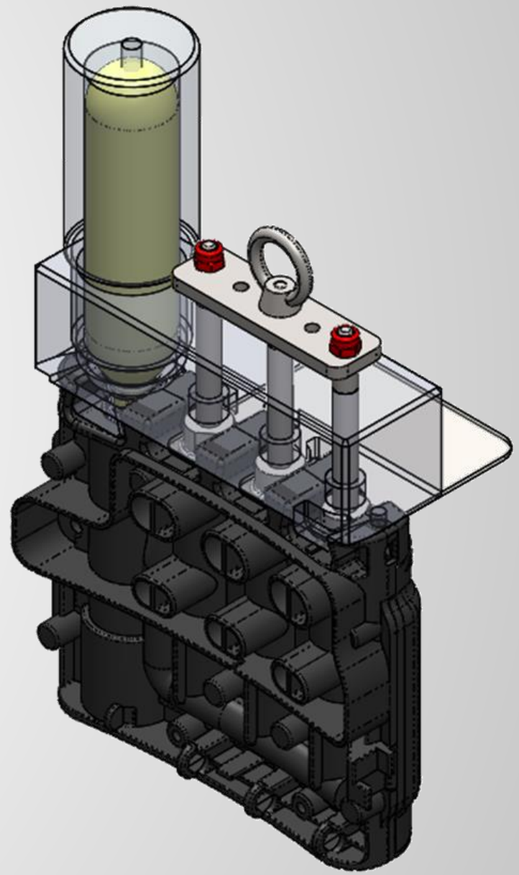
Aluminum protective structure with ballistic glass, designed to withstand gas cartridge impacts (~900 J).Integration of a 45° rotating table or overhead suspension system to ensure repeatable mannequin positioning. Door safety interlock and controlled inflation unit to ensure operator safety and enable reliable data acquisition during testing.

TESTING TOOL

Brief : Develop a validation tool to ensure reliable trigger system performance in after-sales service operations.



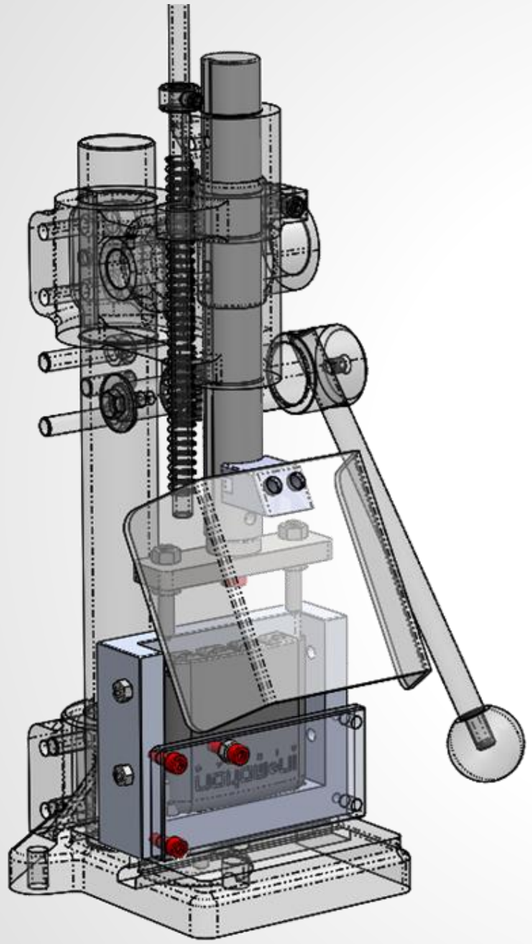




The technician installs a CO_2 cartridge and screws the cover. A pre-cut piece of cardboard is then inserted, and the system is triggered. By inspecting the marked cardboard, it is possible to verify whether the three pistons have correctly reached their impact position.

RECYCLING TOOL

Brief : Develop a tool to separate metal and plastic components for recycling.

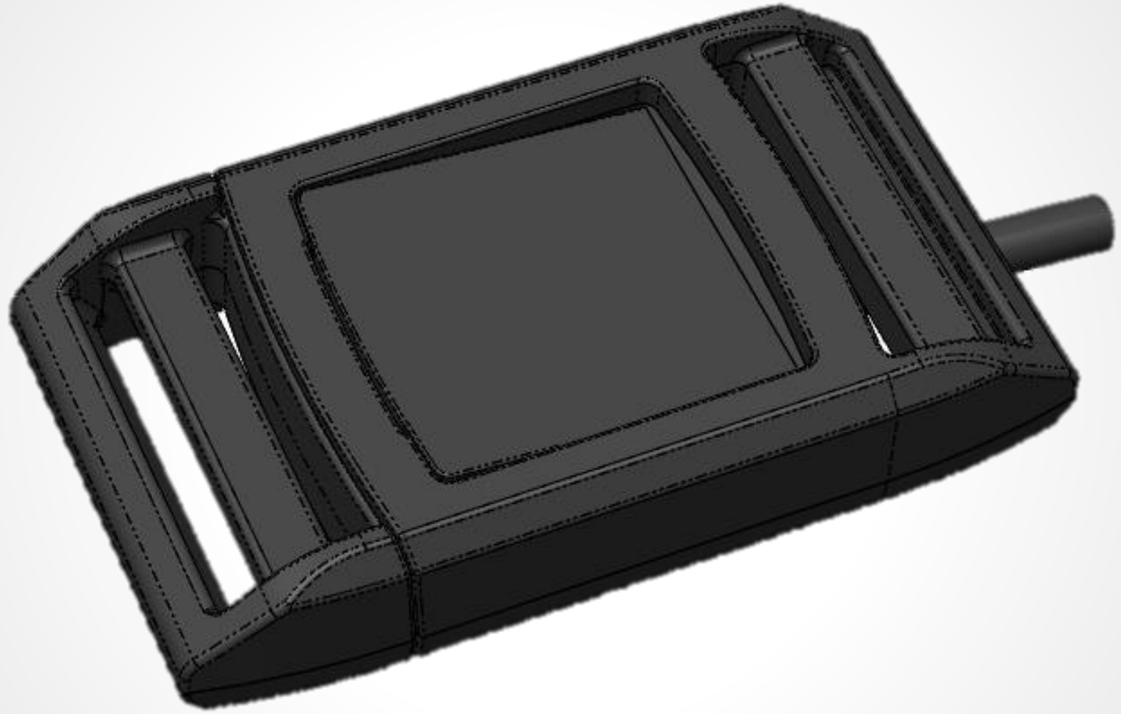




Design of a tool mounted on a basic manual press to separate plastic and metal components of the cartridge, improving recyclability. Prototyping to determine optimal pressure points and ensure operator safety, followed by industrialization and deployment to enable efficient processing by recycling partners.

INNOVATIVE BACKPACK BUCKLE

Brief : Design a buckle capable of withstanding airbag deployment forces, detecting closure using a Hall effect sensor and magnet, and providing user feedback through an integrated LED indicator



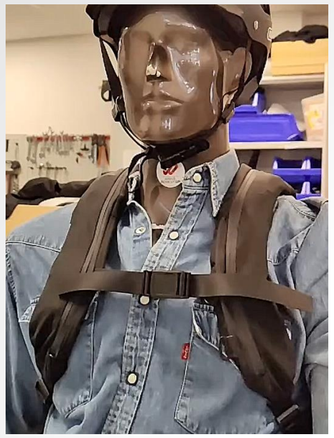


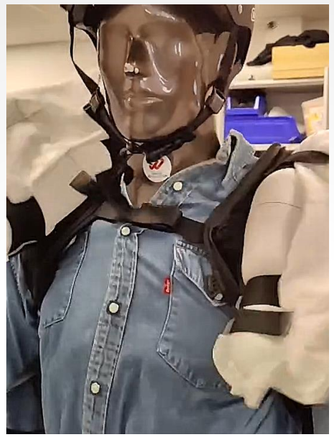


WORK

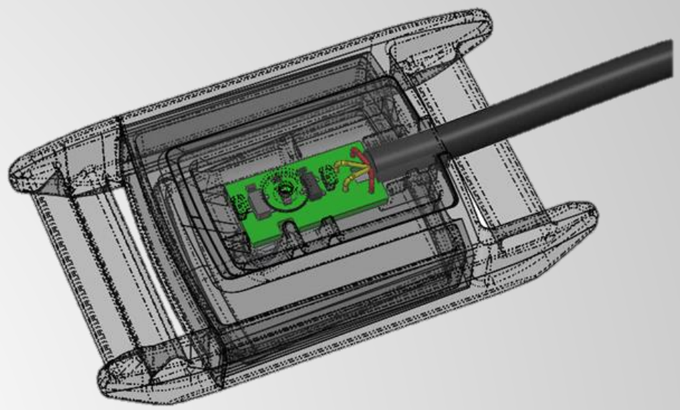
The project involved analyzing and quantifying the forces generated during airbag deployment, followed by stress testing of existing buckle designs to identify their performance limits. The design was developed to meet both technical requirements and design intent.

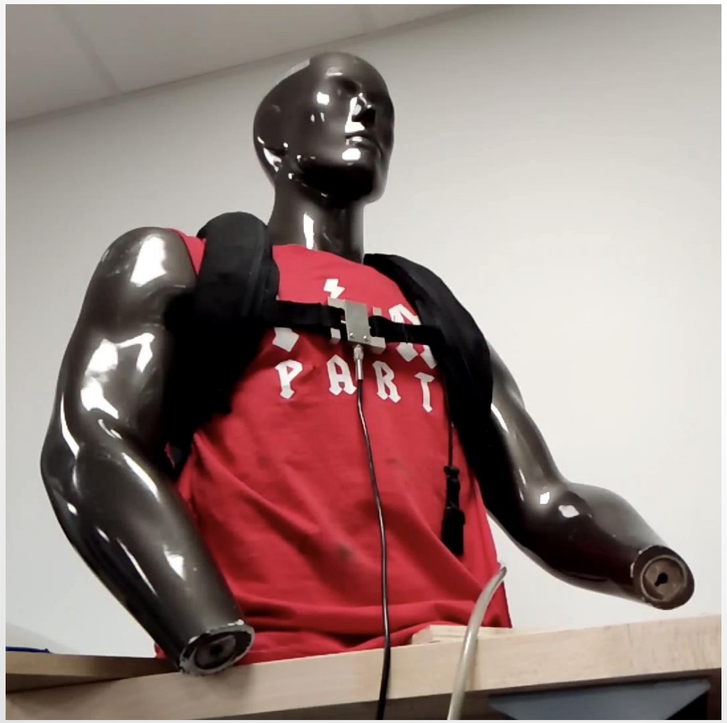
An embedded PCB integrating a Hall effect sensor, magnet, and LED indicator was developed and successfully industrialized through overmolding within the buckle assembly.









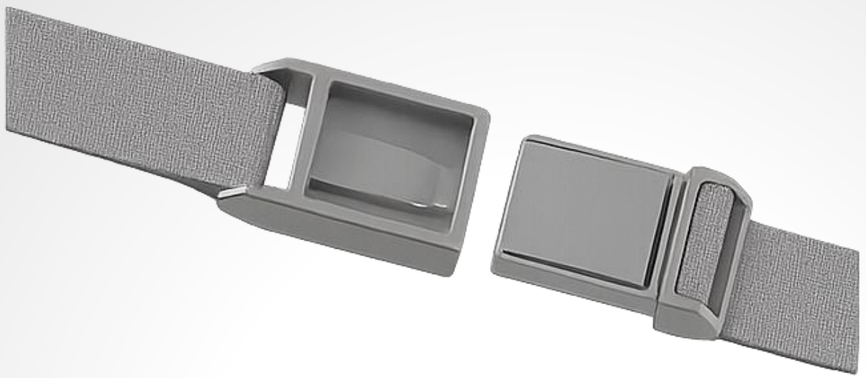


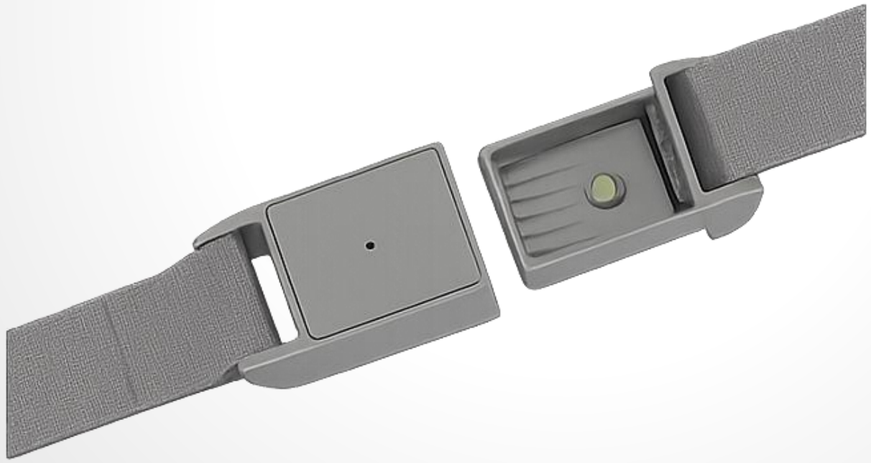


RESULT

The project was completed with a fully functional assembly combining a cable, an overmolded PCB, and a plastic buckle. Nylon was selected for its high mechanical strength and scratch resistance, which are essential for the push-to-release system. The LED provided sufficient brightness for clear user feedback in daylight conditions.

Successful launch of two plastic injection molds and one PCB overmolding mold, including the integration of an optical fiber for LED guidance.





OPEN



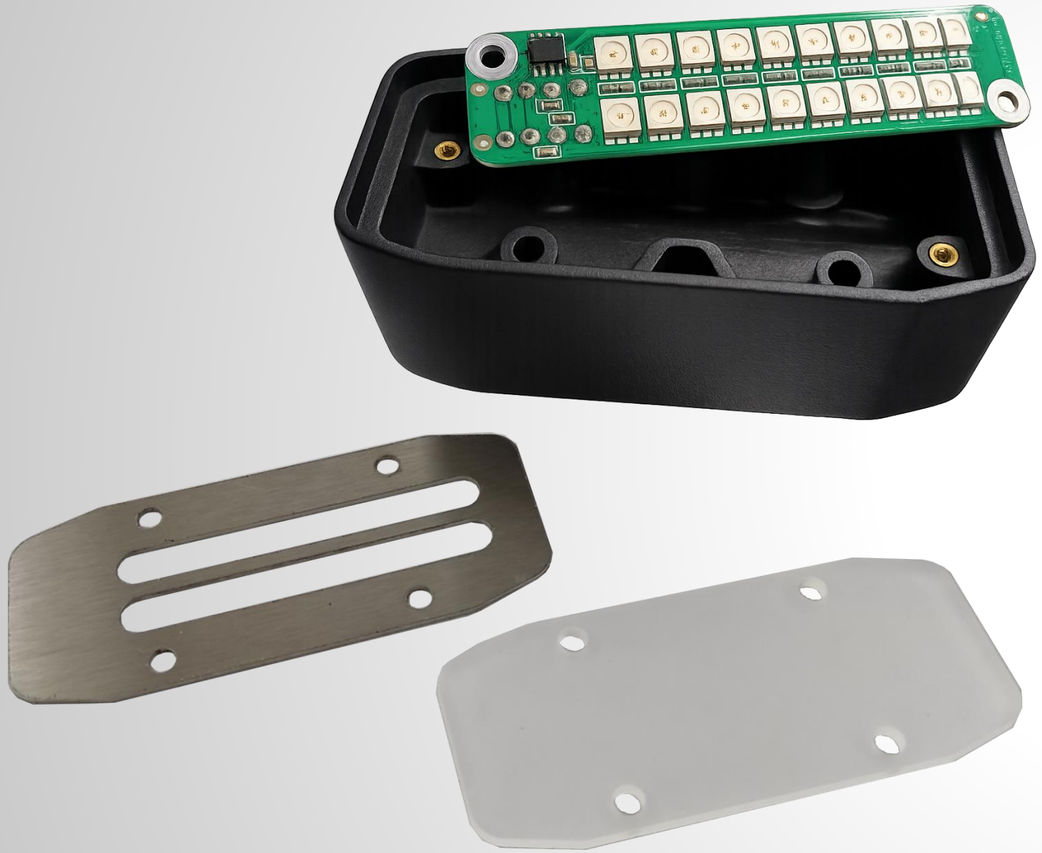


CLOSE

MOONBIKES

SPEEDOMETER

Brief : Design and produce of a speedometer featuring an LED display

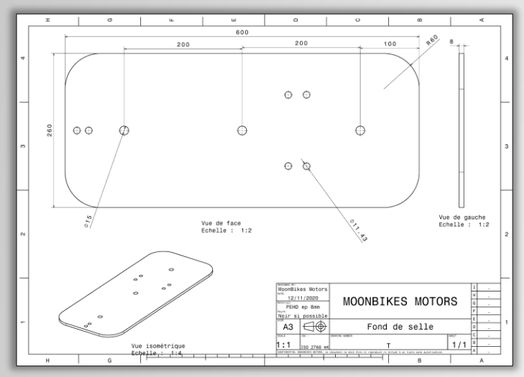


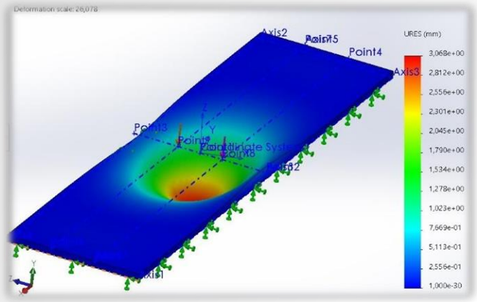


Brushed stainless steel combined with a translucent plastic insert ensures even diffusion of the LED lighting, creating a refined visual effect. The pictograms are laser-engraved onto the metal surface, while the housing is produced using additive manufacturing, making it well suited for a production batch of 50 units.

SEAT

Brief : Create a comfortable seat for two users respecting the simplest design of the machine











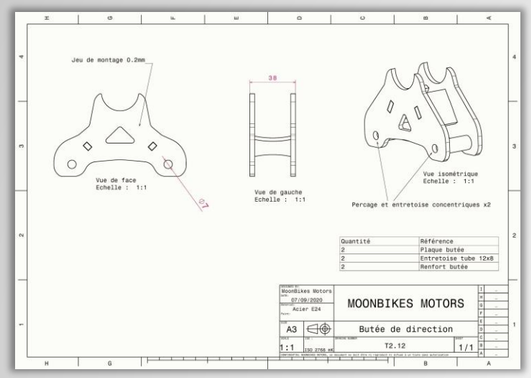


A dual-density foam was selected to enhance comfort, while the HDPE seat base provides structural rigidity and integrates threaded inserts to attach a hinge system, allowing access to the batteries.

MOONBIKES

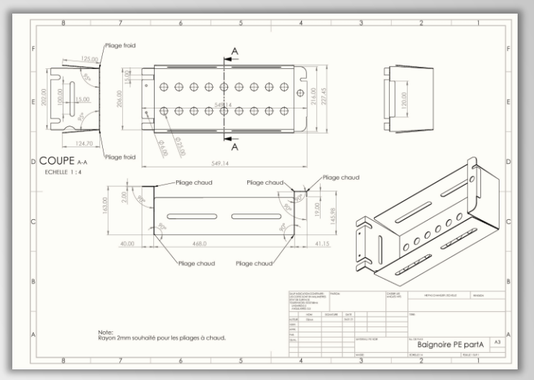
SMALL BUT IMPORTANT

Brief : Improve build quality and user safety



Steering stop

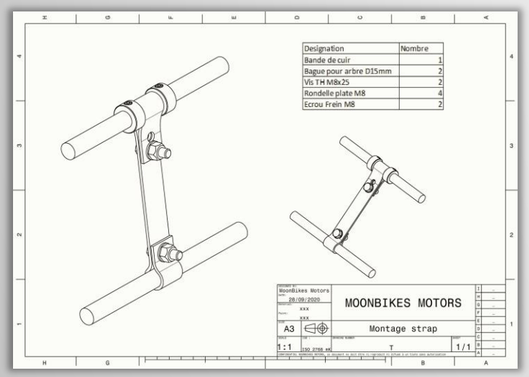
Enhances usability and rider safety while protecting cables from damage.



Electrical protection cover

Secures access to the batteries and high-voltage cables in snowy conditions with a folded plastic design.





Safety strap

Prevents complete tracked drive unit failure in the event of a suspension malfunction.

Nylon track guides

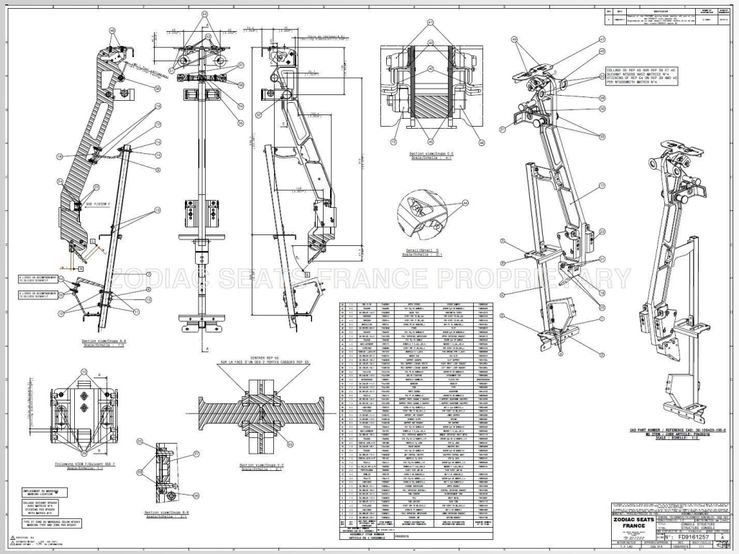
Reduces derailment risk and minimizes friction noise using a low-cost 2D-machined component.



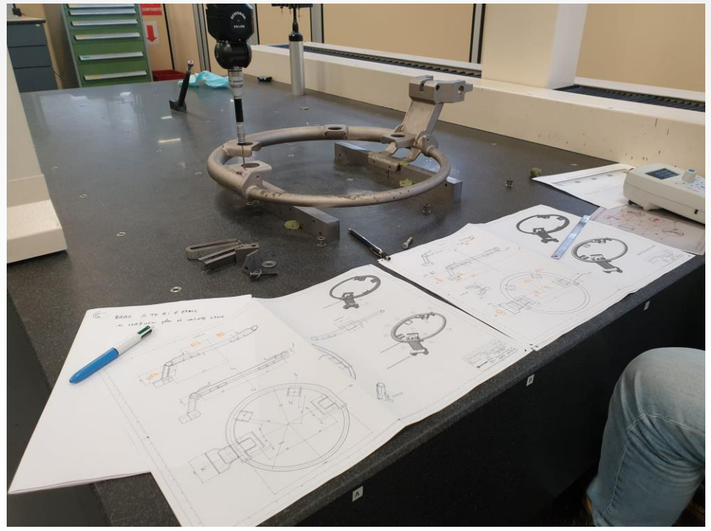
I applied my technical expertise before and during the first production of 50 machines by adding these components to prevent user issues.
I was also in charge of selecting and supplying all basic components, including screws, cables, production tools, and some electrical parts.

LARGE SCALE MEASUREMENT TOOL

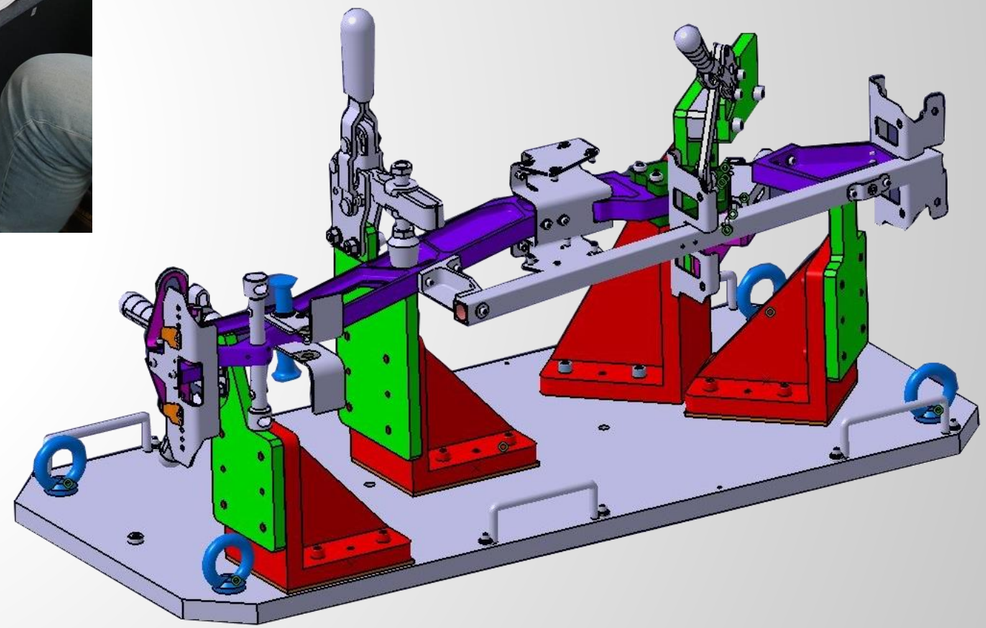
Brief : Manufacture a clamping fixture for CMM inspection of a large metal part



Zodiac boat component to measure



Coordinate Measuring Machine

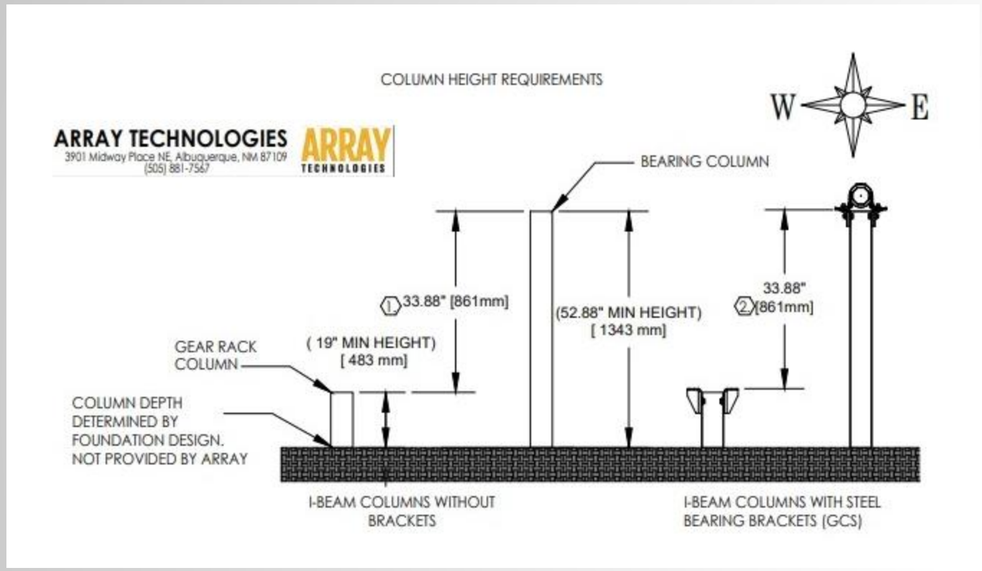


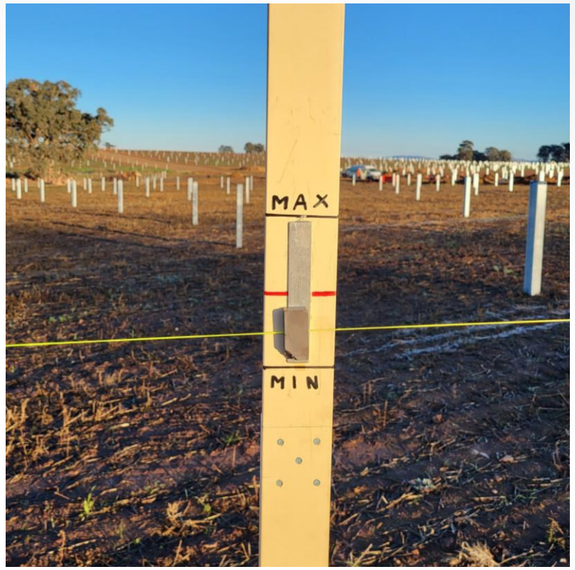
Measurement tool

After a technical call with Zodiac to confirm the measurement plan, an aluminium base with angle brackets and toggle clamps was designed to secure the part without interfering with the probe path.

SOLAR FARM TOOL

Brief : Create a tool to measure the linearity defect of solar panel tubes installed on top of stacks







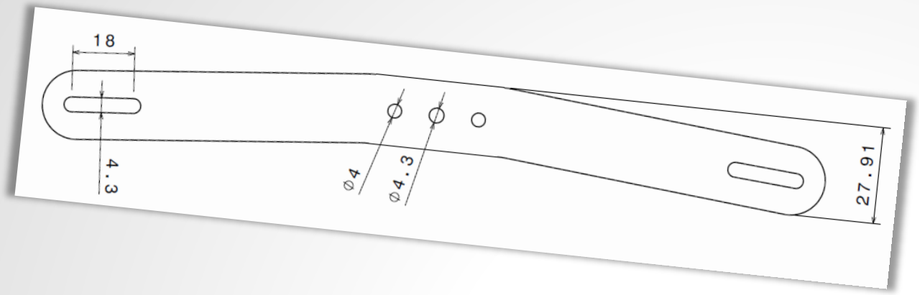
Even though the motor stack is shorter, measurements still rely on it. A dummy stack was created to simulate the motor height and provide a reference point. A magnet holds a mason's line, which is used to measure adjacent stacks and check if adjustments are needed.

This simple tool is critical, as even a small error in stack height can cost at least $1,000 if detected early, and much more after panel installation !

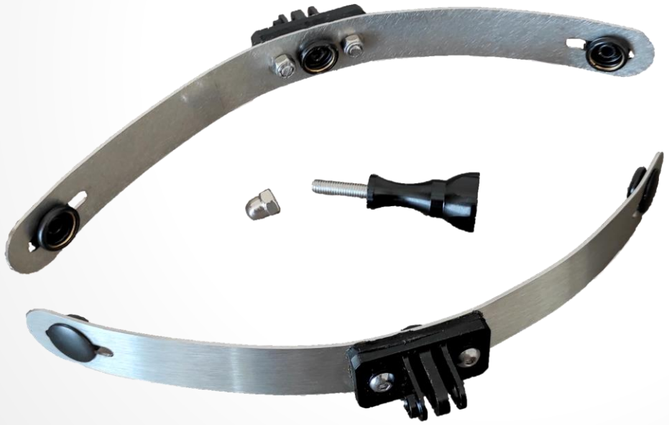
DUKOR
motorcycle

GOPRO SUPPORT FOR VINTAGE HELMET

Brief : Design a plug-in system to mount a GoPro on a vintage helmet without using 3M adhesive









A stainless steel plate is laser-cut and assembled with high-quality snap buttons. The product is bent to match the shape of the helmet, and the external buttons are adjustable to fit all helmet models. Users can install the system in just a few seconds to film their ride.

Product sold for few years

DRIFT TRIKE

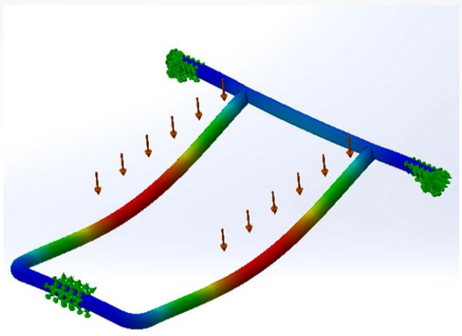
Brief : Creating a fun drift tricycle. The vehicle must be adaptable to all sizes



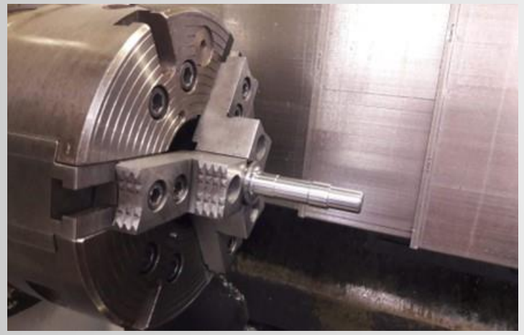
Scanning of the frame



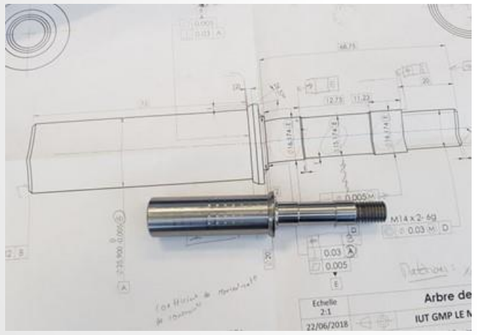
3D modeling of the trike



FDM testing of critical parts



Machining of axles



Dimensional inspection



Laser cutting of the seat base



The Drift Trike project was re-used by next year student to add an electric motor.

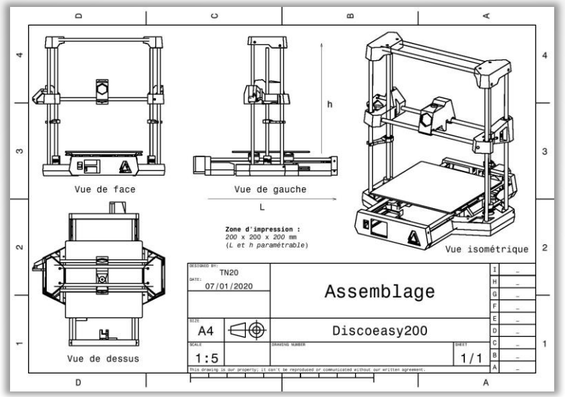
The bicycle frame was 3D scanned and the resulting point cloud processed to create a complete digital model of the trike. Tubes were precisely cut, bent, and welded, while wheel axles were machined and welded to the chassis. The seat adjustment system was laser-cut and assembled, ensuring full functionality and adaptability.

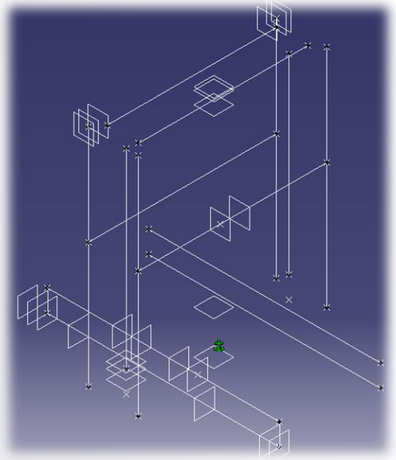
3D PRINTER REVERSE ENGINEERED

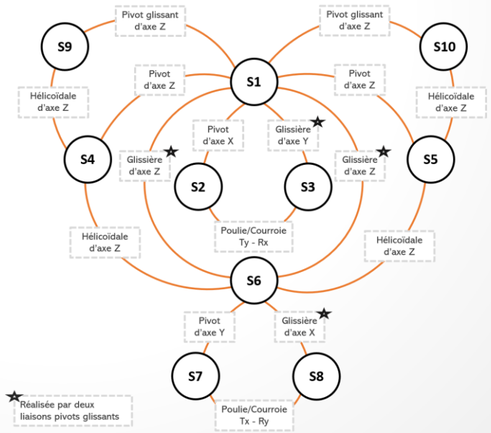
Brief : Measure and re-engineer a 3D printer using skeleton method



Real printer









Modeled printer

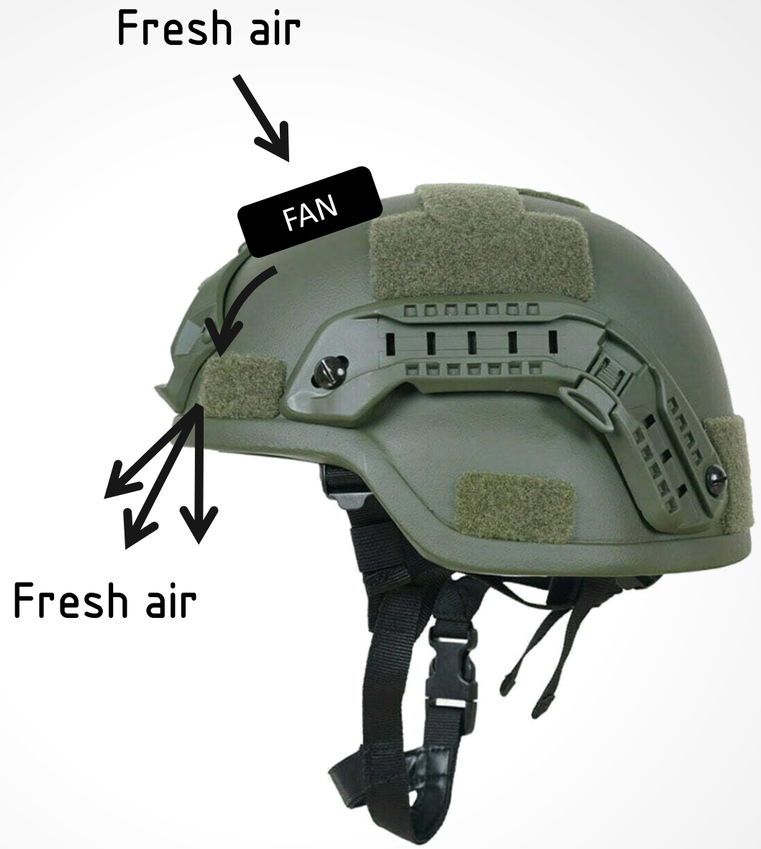
Measure all components using different strategies, reverse-engineer all parts in Creo, create a parametric skeleton in CATIA, and import the .stp files from Creo to build the assembly.

Personal & Academic
PROJECTS

AIRSOFT HELMET ANTI-FOG SYSTEM

Brief : Help airsoft players who suffer from fogging on their goggles by designing a fan system



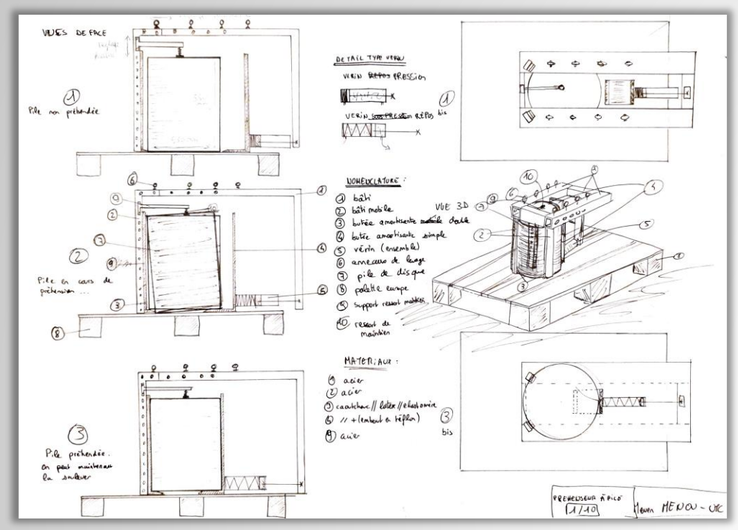


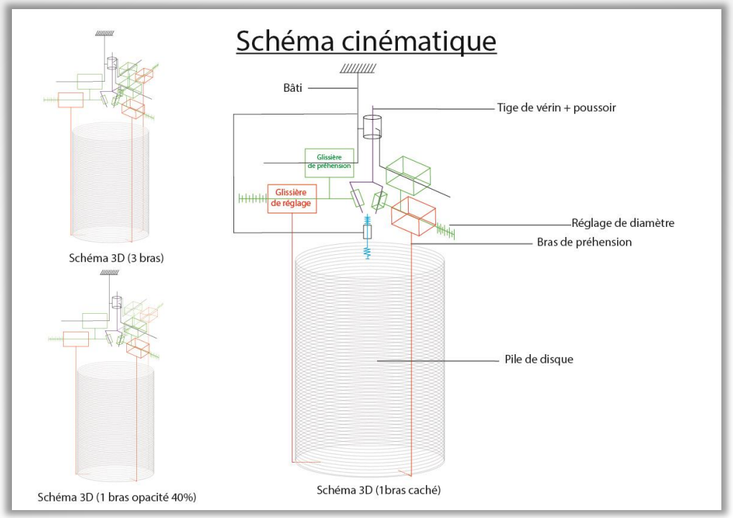


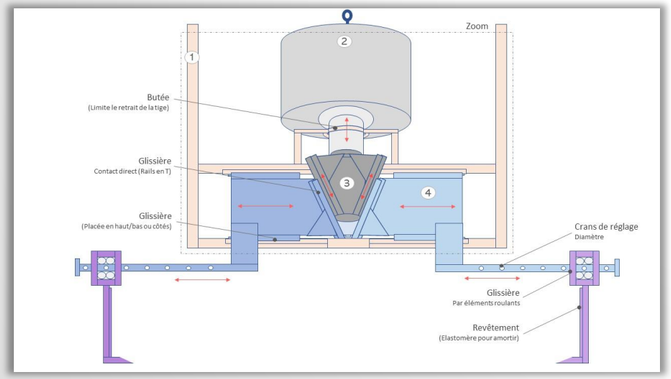
A fan, controlled by a 3-speed switch and powered by a power bank, draws in fresh, dry air and directs it to the goggles via a 3D-printed duct system. Fog disappears within seconds, and the low-speed setting keeps the goggles clear throughout entire games !

SHEET STACK GRIPPER

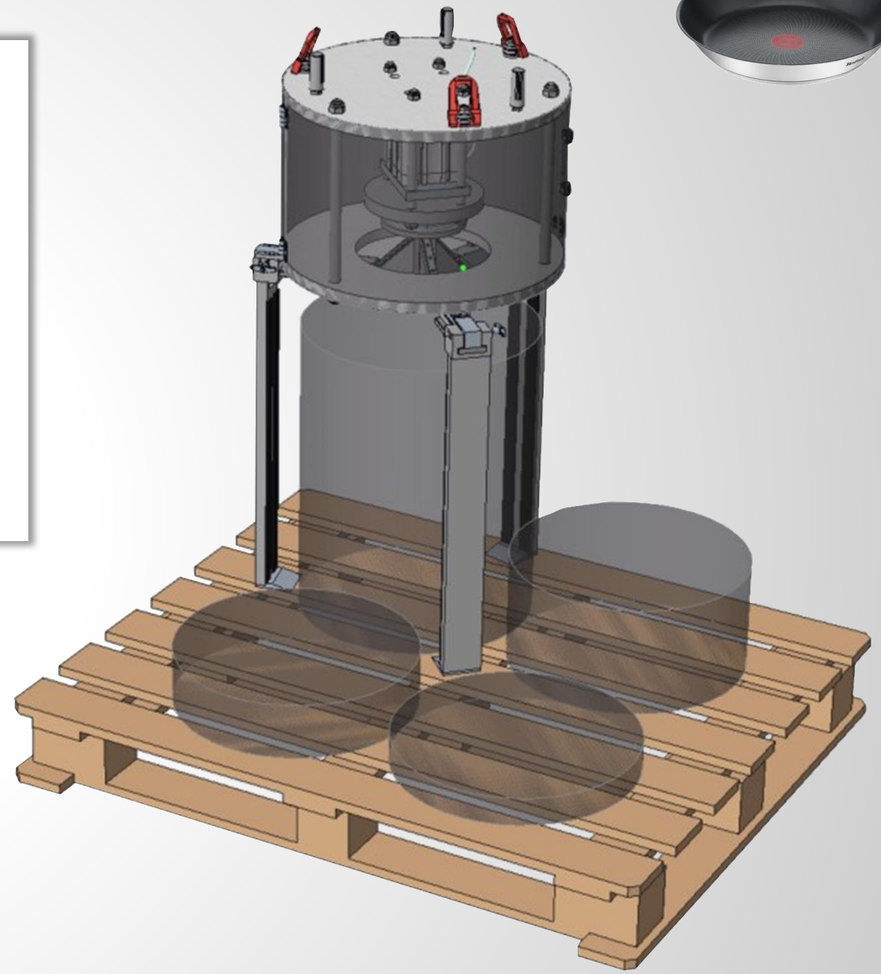
Brief : Design a pneumatic tool capable of lifting stacks of sheets of various diameters







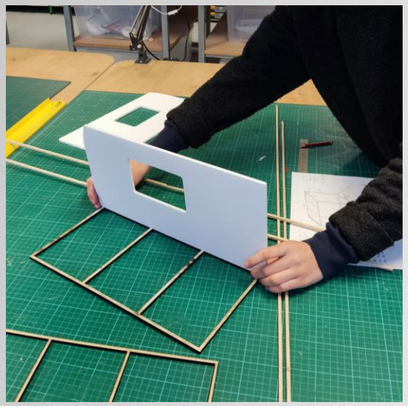




Pneumatic cylinder sizing was based on the weight of the elements, applied mechanics, and trigonometry. The use of standardized components, an efficient anti-friction solution, and manual adjustment allowed adaptation to different stack diameters, resulting in a simple and low-cost design. The thorough study enabled the design to be modeled in Creo (a new software for me) in just two days. The solution was praised by the jury at Tefal, the company that initiated the study.

LIGHT BICYCLE TRAIL

Brief : Design a bicycle trail solution weighing under 100 kg and create a 1:6 scale model



Wooden frame & foam board



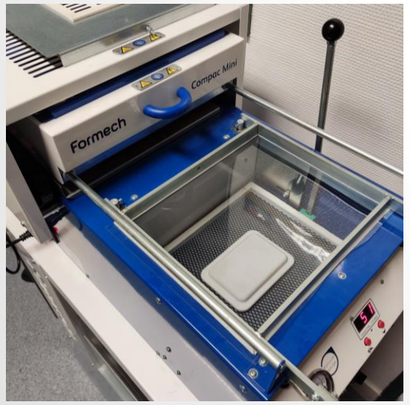
3D modeled in one day



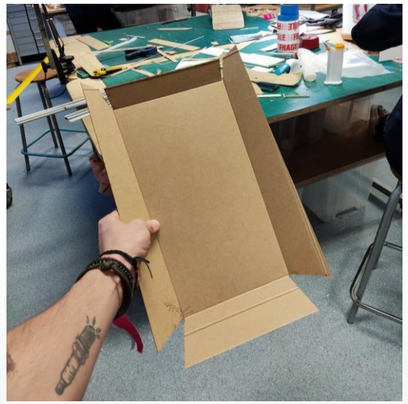
Laser-cut cardboard



3D-printed negative mold



Thermoformed windows



Folding helped by 3D

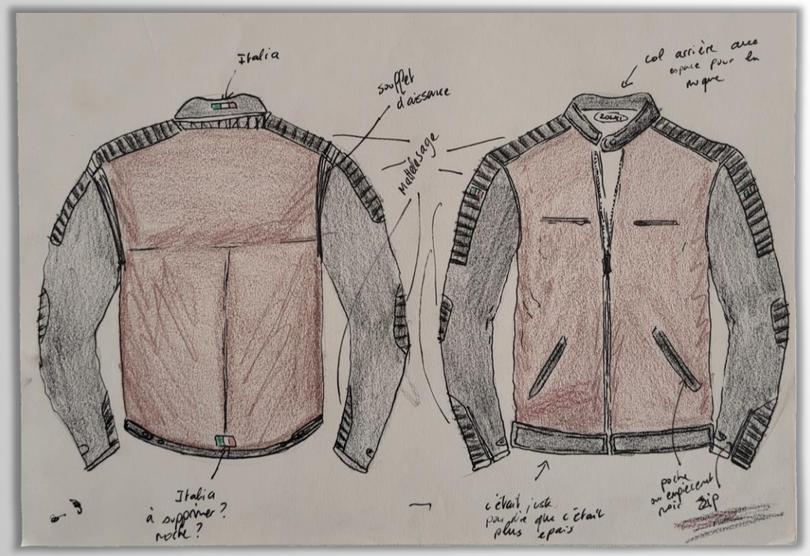


My expertise in 3D modeling allowed me to verify the design specifications (weight and dimensions) and facilitate the prototype construction using 2D plans. Awarded first place by the jury, thanks to the quality of the model.

VINTAGE MOTORCYCLE JACKET

Brief : Design a vintage motorcycle jacket

Partnership with the motorcycle brand Zolki. Designed prototype in collaboration with the Indian factory, then launched into production. Developed features were added to the entire collection (neoprene neck, pant attachments, modified sleeves, etc.).





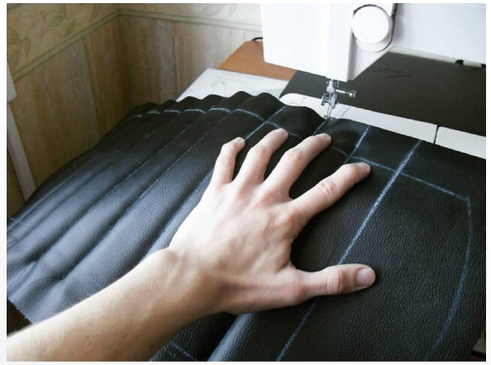


SCRAMBLER 125

Brief : Transform a Yamaha Sr 125 to a scrambler bike









Full mechanical restoration after six years of inactivity and creation of a new seat, replacement of components, modification of the tank shape, and paintwork.

THANK YOU !

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